Systems of Equations Error Analysis TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Substituting an expression

ORIGINAL PROBLEM

$y = 2x - 1$ and $3x + 2y = 12$

INCORRECT WORKED SOLUTION

$3x + 2y = 12$

$3x + 2 \cdot 2x - 1 = 12$

$7x - 1 = 12$

$7x = 13$

$x = \frac{13}{7}$

A. EXPLAIN THE MISTAKE (1 POINT)

When substituting $y = 2x - 1$, the WHOLE expression must go in parentheses: $2(2x - 1)$. The wrong work multiplied only the $2x$.

B. SHOW CORRECT WORK (1 POINT)

$3x + 2(2x - 1) = 12$

$3x + 4x - 2 = 12$

$7x = 14$

$x = 2$, $y = 2(2) - 1 = 3$

Solution: $(2, 3)$

2. Finishing the system

ORIGINAL PROBLEM

$x + y = 7$ and $x - y = 1$

INCORRECT WORKED SOLUTION

Add the equations: $2x = 8$

$x = 4$

Solution: $x = 4$

A. EXPLAIN THE MISTAKE (1 POINT)

A solution to a system is an ordered pair (x, y). After finding $x = 4$, substitute to find y: $4 + y = 7$, so $y = 3$.

B. SHOW CORRECT WORK (1 POINT)

$2x = 8$, so $x = 4$

$4 + y = 7$

$y = 3$

Solution: $(4, 3)$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Subtracting equations

ORIGINAL PROBLEM

5x + 3y = 21 and 2x + 3y = 12

INCORRECT WORKED SOLUTION

Subtract the equations:

3x = 21 + 12

3x = 33

x = 11

A. EXPLAIN THE MISTAKE (1 POINT)

Subtracting the equations means subtracting the right sides too: 21 − 12 = 9, not 21 + 12.

B. SHOW CORRECT WORK (1 POINT)

Subtract the equations:

3x = 21 − 12

3x = 9

x = 3

2(3) + 3y = 12, so y = 2

Solution: (3, 2)

4. Multiplying the whole equation

ORIGINAL PROBLEM

x + 2y = 8 and 3x − y = 3

INCORRECT WORKED SOLUTION

Multiply the second equation by 2:

6x − 2y = 3

Add to the first: 7x = 11

$x = \frac{11}{7}$

A. EXPLAIN THE MISTAKE (1 POINT)

Multiplying an equation by 2 multiplies EVERY term, including the 3 on the right: 6x − 2y = 6.

B. SHOW CORRECT WORK (1 POINT)

Multiply the second equation by 2:

6x − 2y = 6

Add to the first: 7x = 14

x = 2, y = 3

Solution: (2, 3)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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5. Parallel lines

ORIGINAL PROBLEM

$y = 2x + 3$ and $y = 2x - 1$

INCORRECT WORKED SOLUTION

$2x + 3 = 2x - 1$

$3 = -1$

The x terms cancelled, so $x = 0$.

$y = 2(0) + 3 = 3$. Solution: (0, 3)

A. EXPLAIN THE MISTAKE (1 POINT)

The variables cancelled and the result $3 = -1$ is FALSE. That means the lines are parallel and there is NO solution.

B. SHOW CORRECT WORK (1 POINT)

$2x + 3 = 2x - 1$

$3 = -1$ (false)

The lines have the same slope and different intercepts.

No solution

6. Reading the result

ORIGINAL PROBLEM

$2x + 4y = 8$ and $x + 2y = 4$

INCORRECT WORKED SOLUTION

Multiply the second equation by -2:

$-2x - 4y = -8$

Add: $0 = 0$

No solution

A. EXPLAIN THE MISTAKE (1 POINT)

The result $0 = 0$ is always TRUE. Both equations describe the same line, so there are infinitely many solutions.

B. SHOW CORRECT WORK (1 POINT)

Multiply the second equation by -2:

$-2x - 4y = -8$

Add: $0 = 0$ (always true)

Infinitely many solutions

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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7. Checking a solution

ORIGINAL PROBLEM

Is (2, 1) a solution of x + y = 3 and 2x − y = 4?

INCORRECT WORKED SOLUTION

Check x + y = 3: 2 + 1 = 3 ✓

The point works, so (2, 1) is a solution.

A. EXPLAIN THE MISTAKE (1 POINT)

A solution must make BOTH equations true. In the second equation, 2(2) − 1 = 3, not 4.

B. SHOW CORRECT WORK (1 POINT)

x + y = 3: 2 + 1 = 3 ✓

2x − y = 4: 2(2) − 1 = 3, not 4 ✗

(2, 1) is NOT a solution.

8. Setting up equations

ORIGINAL PROBLEM

Adult tickets cost $8 and student tickets cost $5. 40 tickets were sold for $275. Write a system.

INCORRECT WORKED SOLUTION

a = adult tickets, s = student tickets

a + s = 275

8a + 5s = 40

A. EXPLAIN THE MISTAKE (1 POINT)

The numbers were put in the wrong equations. The TOTAL TICKETS (40) goes with a + s, and the TOTAL MONEY ($275) goes with 8a + 5s.

B. SHOW CORRECT WORK (1 POINT)

a = adult tickets, s = student tickets

a + s = 40

8a + 5s = 275

(Solving gives a = 25, s = 15.)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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9. Ordered pair

ORIGINAL PROBLEM

$x + y = 10$ and $x - y = 4$

INCORRECT WORKED SOLUTION

Add: $2x = 14$, so $x = 7$

$7 + y = 10$, so $y = 3$

Solution: (3, 7)

A. EXPLAIN THE MISTAKE (1 POINT)

Ordered pairs are written (x, y). Since $x = 7$ and $y = 3$, the solution is (7, 3).

B. SHOW CORRECT WORK (1 POINT)

$2x = 14$, so $x = 7$

$7 + y = 10$, so $y = 3$

Solution: (7, 3)

10. Combining like terms

ORIGINAL PROBLEM

$x = 3 - y$ and $2x + 3y = 8$

INCORRECT WORKED SOLUTION

$2(3 - y) + 3y = 8$

$6 - 2y + 3y = 8$

$6 + 5y = 8$

$5y = 2$

$y = \frac{2}{5}$

A. EXPLAIN THE MISTAKE (1 POINT)

Combine like terms carefully: $-2y + 3y = y$, not $5y$.

B. SHOW CORRECT WORK (1 POINT)

$2(3 - y) + 3y = 8$

$6 - 2y + 3y = 8$

$6 + y = 8$

$y = 2$

$x = 3 - 2 = 1$

Solution: (1, 2)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points